

Lab: Firewalls

The purpose of this lab is to gain practical experience in configuring firewalls. Based on a specified set of rules, a firewall controls incoming and outgoing network traffic on both hosts and networks. In this lab, you will gain hands-on experience with configuring *iptables*, a powerful command-line tool used to manage the Linux kernel's built-in firewall.

Refer to Table 1 to get started as well as the references [1, 2]. Basic command syntax:

iptables -t <table> <action> <chain> <conditions> -j <target>

Table 1: Selected options for iptables.

table-type	Filter (default) NAT Mangle Raw
action	-I to Insert -A to Append -D to Delete -P to set Policy -L to List -F to Flush
chain	For Filter table: INPUT OUTPUT FORWARD
conditions	-s <source ip> -d <destination ip> -i <eth>in interface -o <eth>out interface -p tcp --dport <num> -p tcp --sport <num> -p udp --dport <num> -p udp --sport <num> -p icmp --icmp-type echo-request -p icmp --icmp-type echo-reply -m state --state NEW,RELATED,ESTABLISHED -m mac --mac-source <mac address> -m limit --limit <rate>
target	For Filter table: ACCEPT DROP REJECT LOG

Lab Environment

As illustrated in Figure 1, the lab environment consists of two LANs connected through a gateway within a virtual network topology. During the lab, you will configure the gateway's firewall to secure the internal network (LAN 1). The login credentials for the respective virtual machines are provided below:

root/root (Kali Linux),

gadmin/gadmin (gateway),

madmin/madmin (metasploitable).

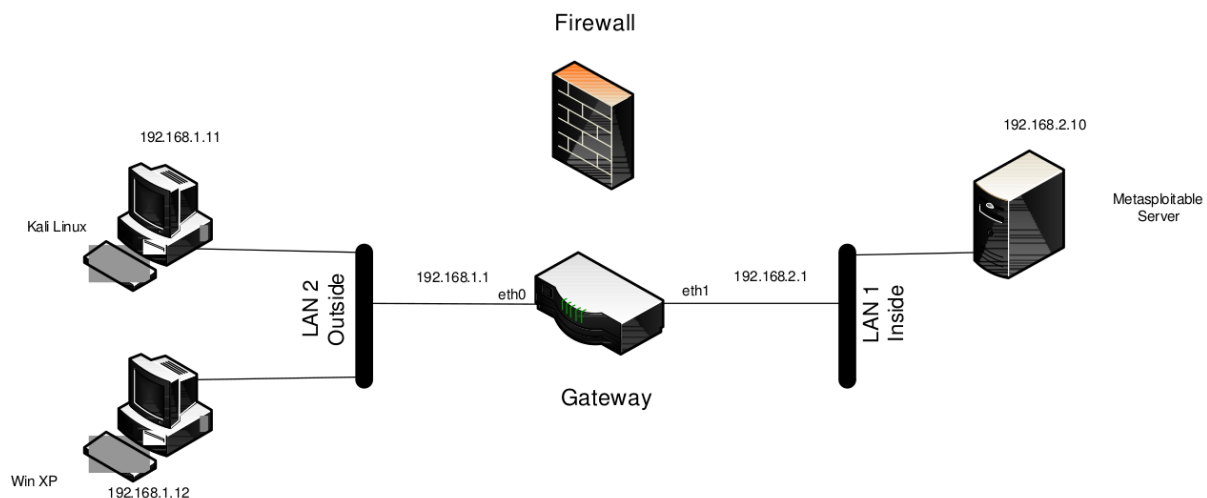


Figure 1: Virtual network topology that is hosted on a single machine.

Exercise 1. Default Policy

1. Before configuring the firewall, identify the purpose of the INPUT, OUTPUT, and FORWARD chains in the filter table. Which type of traffic does each chain handle?
2. Choose a default policy for each of the INPUT, OUTPUT, and FORWARD chains. Explain the reasoning behind your choices and consider the security implications of allowing or denying traffic by default.
3. Based on your chosen default policies, configure rules to perform the following tasks:
 - a) Allow internal servers to access the outside network and receive the corresponding responses.
 - b) Allow the Windows machine to send Telnet traffic to the Metasploitable server. Use at least one MAC address as part of your rule.

Exercise 2. ICMP Packets Filtering

It is reasonable for a client to check whether a server is reachable by sending an ICMP echo request. However, ICMP packets can also be misused for malicious purposes. Using *iptables*, configure the firewall to allow legitimate ICMP echo requests from the outside network to the internal network. Configure the firewall to limit the rate of ICMP echo requests from a single source.

(Optional) Exercise 3. Defense Against Port Scanning

Port scanning can be used as an information-gathering technique (e.g., identify open ports and available services on a host) and may indicate potentially malicious activity. In this task, you will configure a defense against port scanning by using the *recent* module.

Hint: Try to detect and limit repeated connection attempts from the same source IP address.

1. Configure the firewall to keep track of new connection attempts from external hosts.
2. Define a threshold for the number of connection attempts allowed within a given time period.
3. Temporarily block a source that exceeds the defined threshold.

Hint: The *recent* module provides options such as `--set`, `--update`, `--seconds`, and `--hitcount`. Consider how these options can be combined to track repeated connection attempts from the same source.

References

[1] Iptables man page. [iptables\(8\) - Linux man page](#)

[2] Russell, R.: Linux 2.4 packet filtering howto. [Linux 2.4 Packet Filtering HOWTO: Using iptables](#)